Mechanical Engineering Department, CoE, KNUST

ME 161 Basic Mechanics

Unit 3:

Equilibrium of Rigid Bodies

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INTRODUCTION

Equilibrium Conditions for a Particle

A **particle is in equilibrium** if

- it is **at rest relative to an initial reference frame**
- the body **moves with constant velocity along a straight line** relative to an initial frame

The conditions are :

$$\sum F_x = 0 \qquad \sum F_y = 0 \qquad \sum F_z = 0$$

INTRODUCTION

- A **rigid body** is said to be in **equilibrium** if **both** the **resultant force and resultant moment** acting on it are **zero**.
- **Body** will then **show no tendency to rotate** and it will either **remain at rest** or **move in a straight line with constant velocity**
- This unit deals with the algebraic equations that must be satisfied for a rigid body to be in equilibrium under the action of various types of force systems
- The equations are a means of determining some of the unknown forces acting on a rigid body that is in equilibrium
- Another **important subject** of this unit is the **drawing of free-body diagrams** which is a **way of identifying and taking account of all relevant forces acting on a body** in any given situation

Conditions for Rigid-body Equilibrium

- **Consider the rigid body shown in Fig. 3.1 (a),** which is **in equilibrium under the action of external forces F1, F2, etc**.
- The **ith particle of the rigid body is isolated from the rest of the body** and **all forces acting on it are shown Fig. 3.1 (b)**
- **The forces are of 2 types:**
- **(1)the internal forces which are exerted by all other particles on the isolated particle** (**sum of all internal forces is denoted by fi**
- (2) The **external forces arising, for instance, from gravitational or magnetic forces exerted on the particle by other bodies outside the rigid body** (resultant of all **external forces is denoted by Fi**)

Conditions for Rigid-body Equilibrium

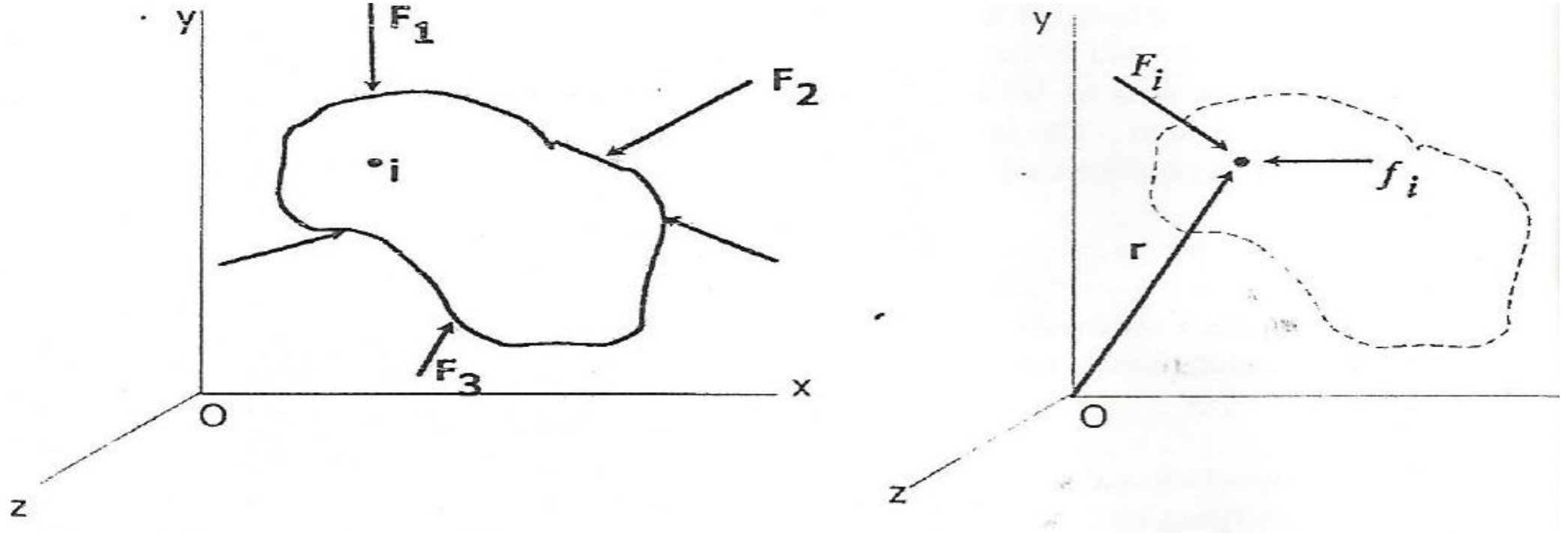


Fig. 3.1Forces acting on the particles of a rigid body

Since the ith particle is in equilibrium , the vector sum of the forces acting on it is zero, i.e

$$f_i + F_i = 0 \qquad (3.1)$$

Similar equations can be written for each of the particles that make up the rigid body. Adding all the equation, we obtain

$$\sum f_i + \sum F_i = 0 \qquad (3.2)$$

Conditions for Rigid-body Equilibrium

The sum $\sum f_i$ of the **internal forces will equal zero since, according to Newton's 3rd Law, the internal forces between particles will occur in equal and opposite pairs**. Therefore, equation (3.2) becomes

$$\sum F_i = 0 \qquad (3.3)$$

The **moments of the forces acting on the ith particle about the arbitrary point O**, Fig 3.1(b), is

$$(r_i \times f_i) + (r_i \times F_i) = 0 \qquad (3.5)$$

We may write similar equations for all other particles and sum them to obtain

$$\sum (r_i \times f_i) + \sum (r_i \times F_i) = 0 \qquad (3.6)$$

The 1st term $\sum (r_i \times f_i)$ is zero since the internal forces occur in equal and opposite pairs
Consequently, equ.(3.6) becomes

$$\sum (r_i \times F_i) = 0$$

or

$$\sum M_o = 0 \qquad (3.7)$$

Conditions for Rigid-body Equilibrium

- **Equations (3.3) and (3.7) are the necessary and sufficient conditions for a rigid body to be in equilibrium**. They require that the **sum of all the external forces acting on the body be equal to zero**, and the **sum of the moments of all external forces about any arbitrary point be equal to zero.**

Free body diagrams

- **Mechanics problems usually involve several bodies interacting together** in one way or the other. In analyzing problems concerning the application of Newton's Laws of motion, it is essential to **identify and take account of all the forces acting on the body**
- It is also important to **exclude any forces that are not directly applied to the body**. One way of ensuring this is to draw a free-body diagram

Steps to follow:

- **Decide which body (or group of bodies) is under consideration and imagine it to be isolated from all other bodies**. **Sketch the outline shape of the body (Step 1)**
- **Indicate, by means of arrows, all external forces and couples that act on the body**. These should include (a) the **weight** of the body (b) all **externally applied loads** (c) **reactions at the supports** from which the body was detached, and (d) **forces due to contact with other bodies from which the body was isolated (Step 2)**
- **Indicate on the diagram those dimensions of the body that would be required for calculating moments of forces**. **Known forces and couples should be labelled with the appropriate magnitudes and directions**. **Use a letter and an arrow to represent the magnitude and directions of each unknown force. An assumed direction can be used and the correct sense determined during calculations.**

Free body diagrams

- To illustrate the procedure, **consider the simple example of blocks of wood, A and B, resting on the ground, one on top of the other**; and the free-body diagram (see Fig.3.2a and b)
- The weights of the blocks are **W_A and W_B** shown acting downwards
- **R_A is the force exerted on block A by block B,** and **R_B the reaction force exerted by block A on block B**
- According to Newton's 3rd Law, **R_A and R_B are equal in magnitude but opposite in sense.**
- **Let R_C be the force exerted on the block B by the ground**

Applying the conditions of equilibrium:

$$W_A - R_A = 0$$

and

$$W_B + R_B - R_C = 0$$

And by Newton's 3rd law:

$$R_A = R_B$$

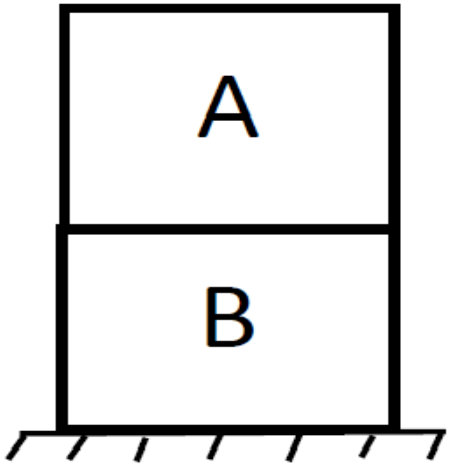
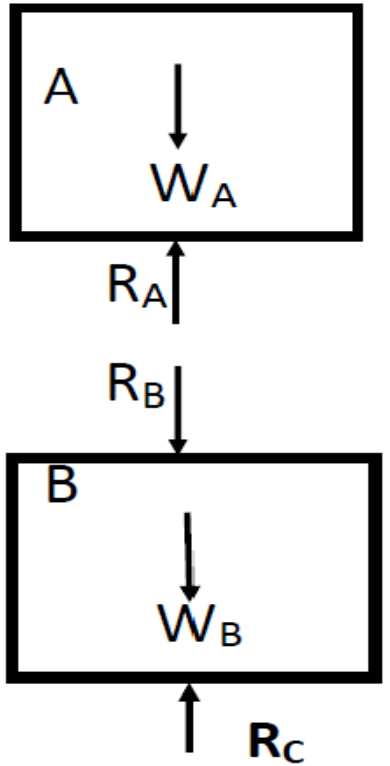


Fig. 3.2 (a)



(b)

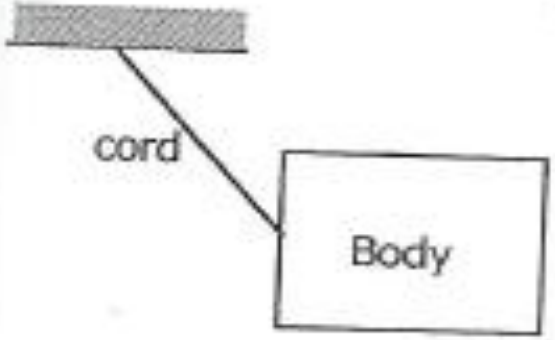
These are 3 equations and 3 unknowns which can easily be solved to obtain:

$$R_C = W_A + W_B$$

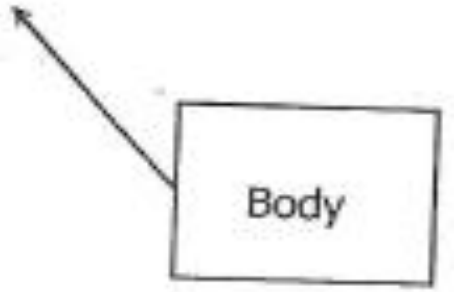
Free body diagrams

- **Practical engineering problems** usually involve bodies connected together in a number of **standard ways**
- Fig.3.3* shows a number of these standard connections between bodies and the ways in which their effects can be represented on the free-body diagram

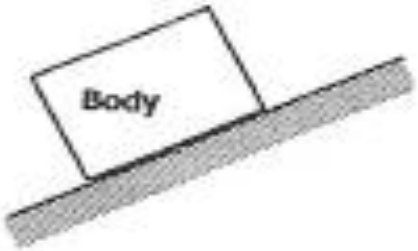
(1) Taut flexible cord, rope or cable



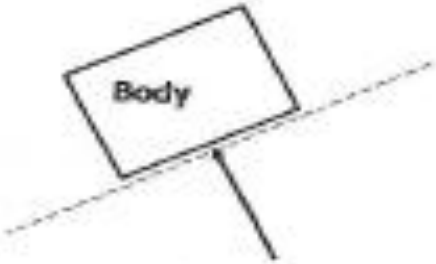
Tensile force along the cord



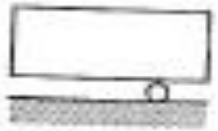
(2) Smooth (frictionless) contacting surface



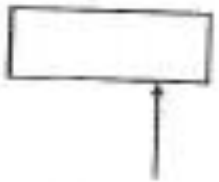
A force which acts perpendicular to the surface at the point of contact



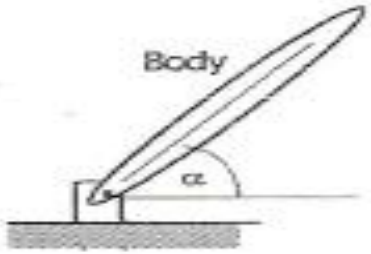
(3) Roller



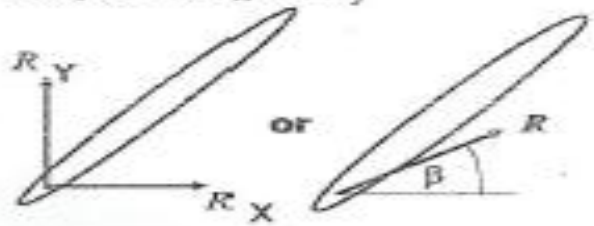
Force perpendicular to the surface on which the roller can roll



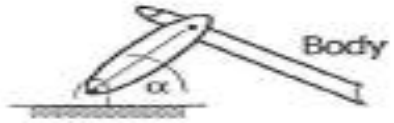
(4) Smooth pin connection or hinge



A force through the pin at an unknown angle, this may be resolved into two components R_x and R_y



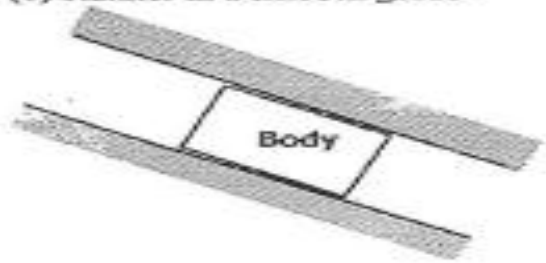
(5) Weightless link



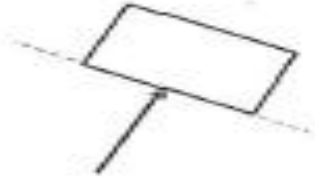
A force which acts along axis of the link



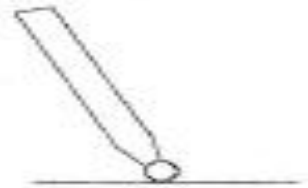
(6) Runner in a smooth guide



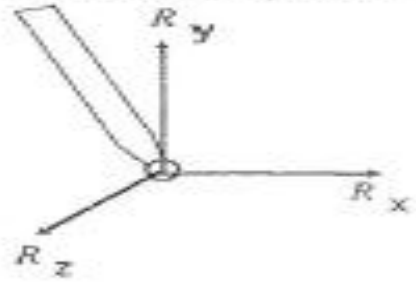
Force normal to the guide



(7) Ball-and-socket



A force at the ball at an unknown angle this force may be resolved into three components R_x, R_y and R_z



Equilibrium in 2-dimensions

For a rigid body to be in a equilibrium, $\sum \mathrm{F} = 0$ and $\sum \mathrm{M_O} = \sum (\mathrm{r} \times \mathrm{F}) = 0$

Resolving the forces and moments into orthogonal components:

$$\sum F_x = 0 \qquad \sum F_y = 0 \qquad \sum F_z = 0$$
$$\sum M_x = 0 \qquad \sum M_y = 0 \qquad \sum M_z = 0$$

When the body is subjected to a system of forces which all lie in the x-y plane, we have

$$\sum F_x = 0 \qquad \sum F_y = 0 \qquad \sum M_z = \sum M_o = 0 \qquad (3.8)$$

The last term may be written as

$$\sum M_A = 0 \qquad and \qquad \sum M_B = 0 \qquad (3.9)$$

Where points A and B lie in the plane of the forces and line AB is not perpendicular to the x-axis

Example 3.1

Determine the horizontal and vertical components of the reactions at A and C for the beam shown in Fig. 3.3(a)

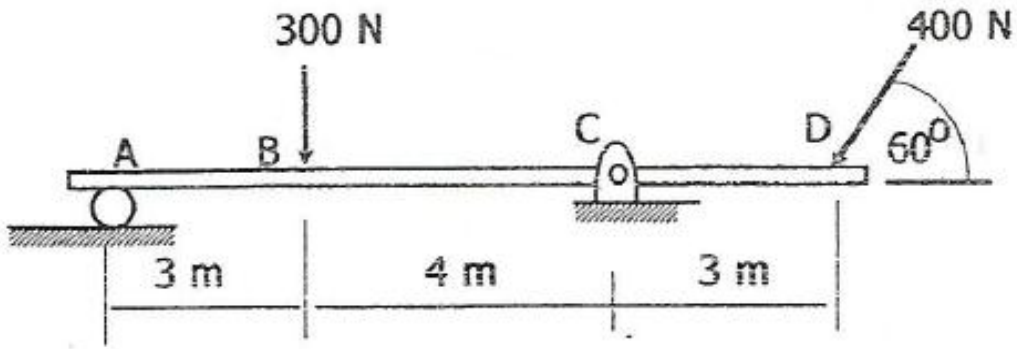


Fig. 3.3 (a)

Solutions

Summing forces in the x-direction,

$$\sum F_x = 0 \Rightarrow C_x - 400\cos 60 = 0$$

$$\therefore C_x = 200\,N$$

Summing forces in the y-direction,

$$\sum F_y = 0 \Rightarrow = A_y - 300 + C_y - 400\sin 60 = 0$$

$$or\ A_y + C_y = 646.4\,N$$

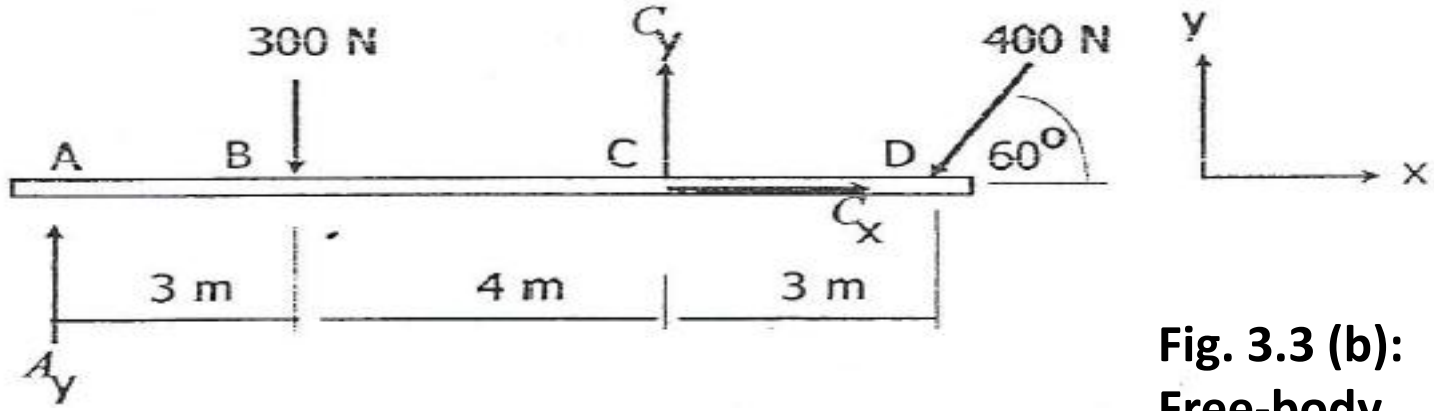


Fig. 3.3 (b): Free-body diagram

Applying the moment equation about the point C,

$$\sum M_C = 0 \Rightarrow 300\times 4 - A_y \times 7 - \left(400\sin 60\right)\times 3 = 0$$

$$\Rightarrow A_y = \frac{300\times 4 - \left(400\sin 60\right)\times 3}{7} = 23\,N$$

$$and\ C_y = 646.4 - A_y = 646.4 - 23.0 = 623.4\,N$$

Example 3.2

In Fig. 3.4 the **lever AC is subjected to a vertical force of 600 N at C**. Determine the **tension in the cable AD and the reaction at B.**

SOLUTION

Taking moments about the point B, we have

$$T\cos 15^o \times 0.4 = 600 \times 0.3\cos 30^o \Rightarrow$$

$$T = \frac{600 \times 0.3\cos 30^o}{0.4 \times \cos 15^o} = 403.5\,N$$

Summing forces in the x-direction, we have

$$T\cos 45^o - B_x = 0 \Rightarrow$$

$$B_x = T\cos 45^o = 403.5\cos 45^o = 285.3\,N$$

Summing forces in the vertical-direction, we have

$$T\sin 45^o - B_y + 600 = 0 \Rightarrow$$

$$B_y = 600 + 403.5\cos 45^o = 885.3\,N$$

The magnitude of the reaction at B is

$$B = \sqrt{B_x^2 + B_y^2} = \sqrt{(285.3)^2 + (885.3)^2} \Rightarrow$$

$$B = 885.3\ N$$

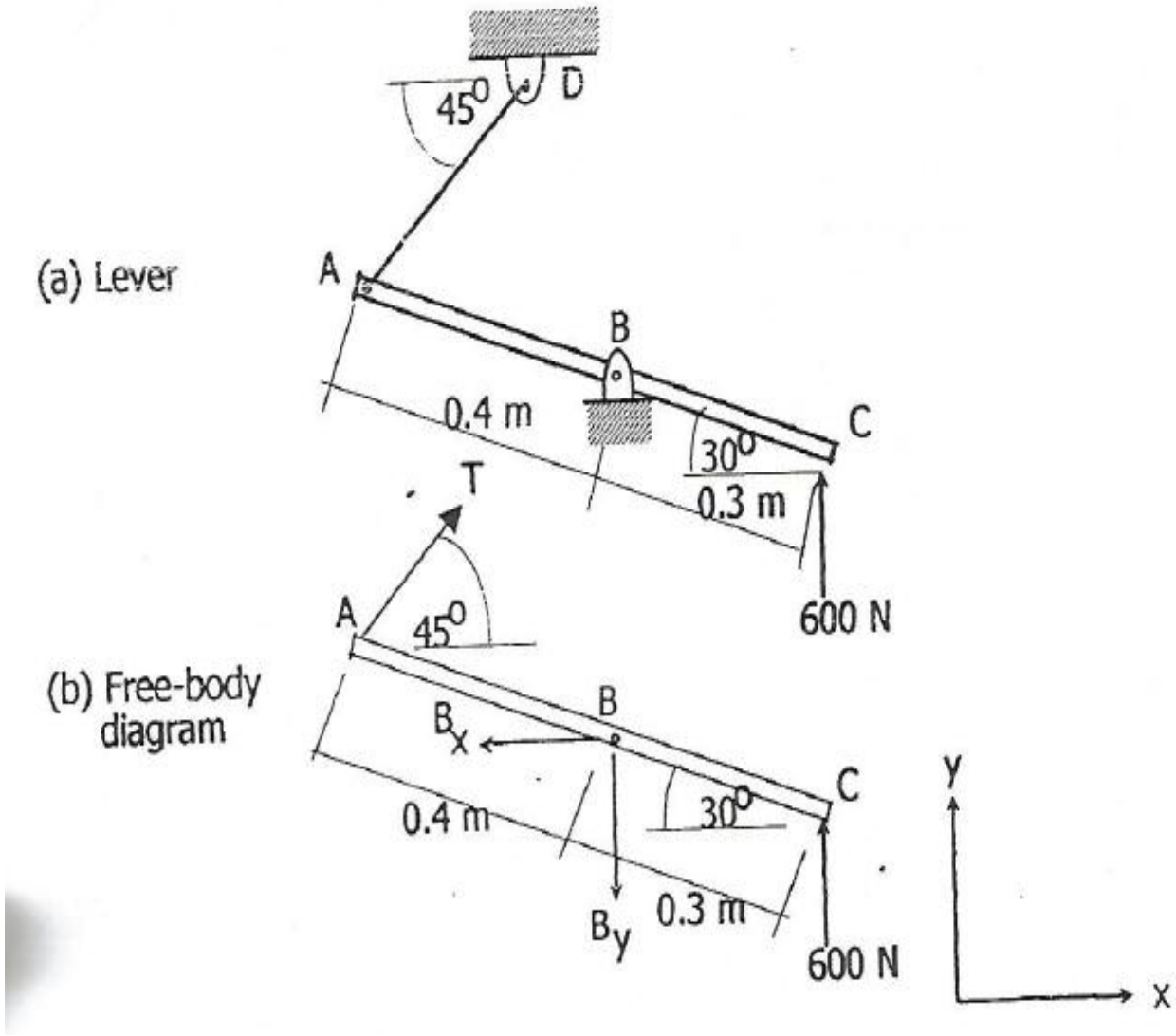


Fig. 3.4

Equilibrium of Two-Force and Three-Force Bodies

- A **body acted** on by only 2 forces is referred to as a 2-force body
- Similarly, **a 3-force body is one which is subjected to 3, and only 3, nonparallel forces**
- The solution of equilibrium problem is very often simplified if we are able to identify the 2-and 3-force members

2-Force Members

- If a **body is held in equilibrium by 2 forces, the forces must be collinear, equal in magnitude and opposite in sense**
- From Fig. 3.5, it is obvious that **if the 2 forces are not equal in magnitude and opposite in sense, the equilibrium conditions $\sum F_x = 0$ and $\sum F_y = 0$ would not be satisfied**
- Also, if the **forces are not collinear**, **the condition $\sum M = 0$ would not be satisfied** since an unbalanced couple moment would exist
- A **more general definition of a 2-force body** is one which is subjected to **forces acting at only 2 points**
- If several forces act at 2 different points on a rigid body, the forces acting at each point can be replaced by their resultant
- **For equilibrium, the 2 resultant forces must be collinear, equal in magnitude and opposite in sense**
- Recognition of 2-force members enables the simplification of certain problems especially in structural analysis

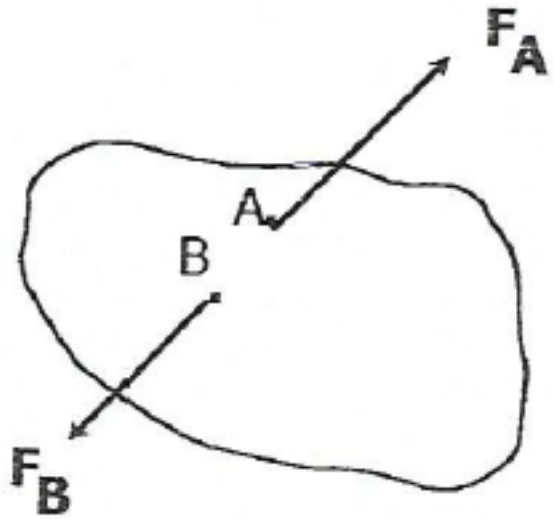
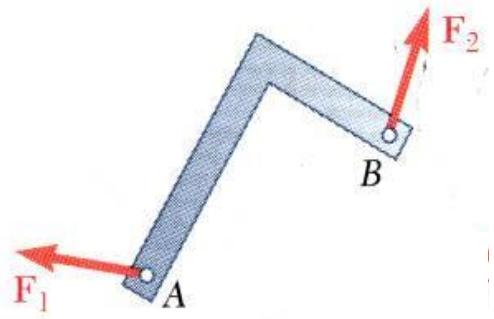
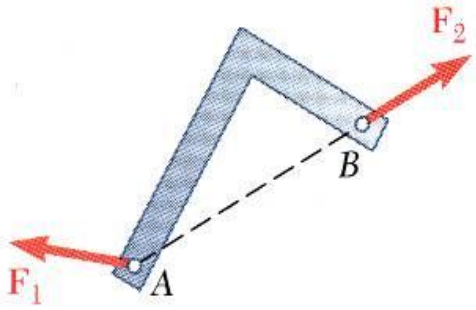
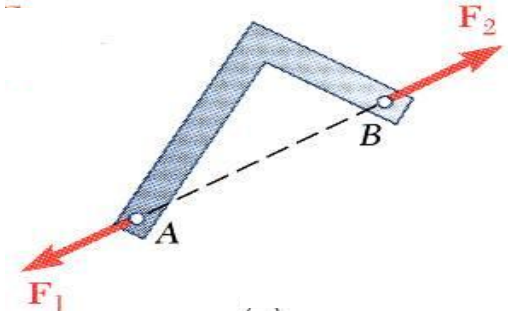


Fig. 3.5

Equilibrium of a Two-Force Body







- Consider a plate subjected to two forces $\boldsymbol{F_1}$ and $\boldsymbol{F_2}$

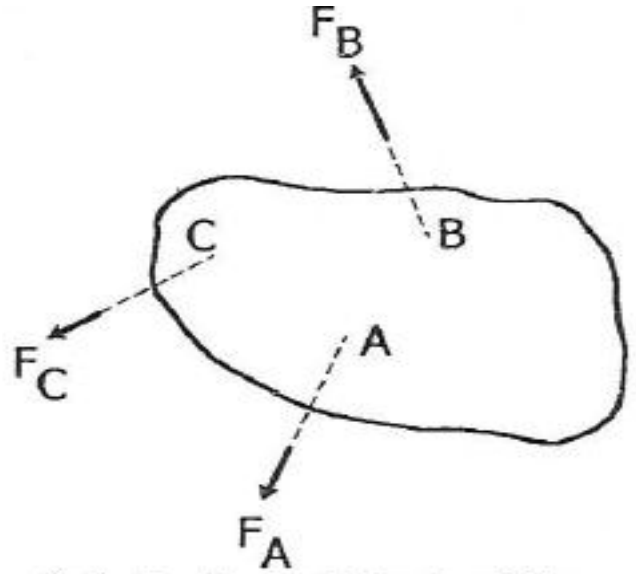
- For static equilibrium, the sum of moments about ***A*** must be zero. The moment of $\boldsymbol{F_2}$ must be zero. It follows that the line of action of $\boldsymbol{F_2}$ must pass through *A*.

- **Similarly, the line of action of $\boldsymbol{F_1}$ must pass through *B* for the sum of moments about *B* to be zero.**

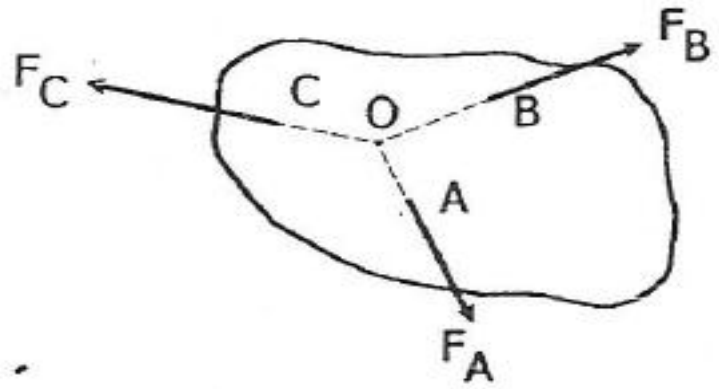
- Requiring that the sum of forces in any direction be zero leads to the conclusion that $\boldsymbol{F_1}$ and $\boldsymbol{F_2}$ must have equal magnitude but opposite sense.

Three-force bodies

- When a **body is held in equilibrium by 3 forces, they must be either concurrent and coplanar, or parallel**.
- Fig. 3.6 (a) shows a rigid body subjected to 3 **forces F_A, F_B and F_C**
- The **orientation of these forces cannot be chosen arbitrarily if the body is to be held in equilibrium**
- The **forces must all lie in the plane ABC or else the conditions $\sum M_x = 0$, $\sum M_y = 0$ and $\sum M_z = 0$ would not be satisfied**
- Now, **let the lines of action of F_A, F_B and F_C intersect at O.** Then the **moments of F_A ,F_B and Fc about O is zero (see Fig. 3.6(b))**
- **In the exceptional case where 2 of the forces are parallel, the 3rd force must also be parallel to the other two forces**



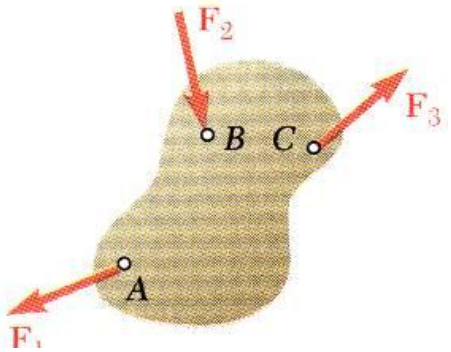
(a) Body subjected to three forces

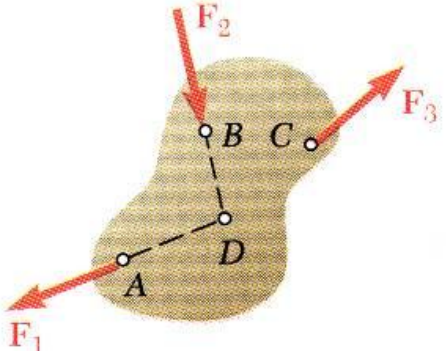


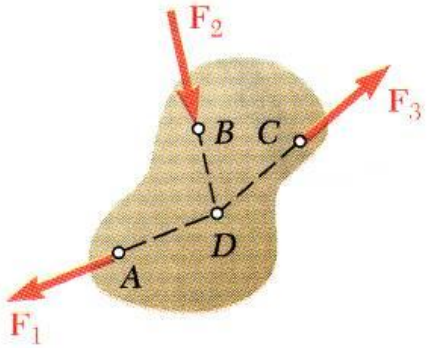
(b) Directions of forces for equilibrium to be maintained

Fig. 3.6 A 3-force body

Equilibrium of a Three-Force Body







- Consider a rigid body subjected to forces acting at only 3 points.
- Assuming that their lines of action intersect, the moment of $\boldsymbol{F_1}$ and $\boldsymbol{F_2}$ about the point of intersection represented by *D* is zero.
- Since the rigid body is in equilibrium, the sum of the moments of $\boldsymbol{F_1}$, $\boldsymbol{F_2}$, and $\boldsymbol{F_3}$ about any axis must be zero. It follows that the moment of $\boldsymbol{F_3}$ about *D* must be zero as well and that the line of action of $\boldsymbol{F_3}$ must pass through *D*.
- The lines of action of the three forces must be concurrent or parallel.

Example 3.3

Solve the problem in example 3.2 by considering the lever as a 3-force body

SOLUTION

The **free-body diagram of the lever is shown in Fig. 3.7** with the components of the reaction at B combined into one force **R acting at an angle θ to the vertical**

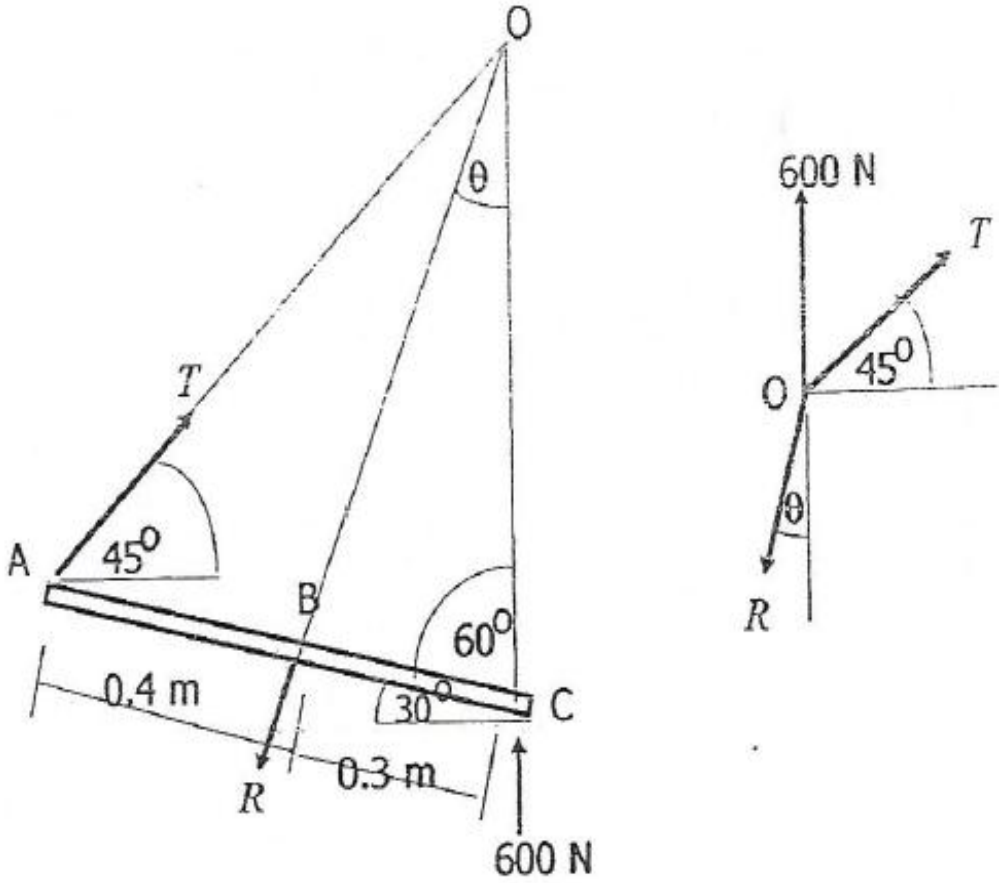


Fig. 3.7

$$\overline{OA} = 0.7\times\left(\frac{\sin 60^o}{\sin 45^o}\right) = 0.8573\,\text{m};\quad \overline{OC} = 0.7\times\left(\frac{\sin 75^o}{\sin 45^o}\right) = 0.9562\,\text{m}$$

$$\overline{OB} = \sqrt{(0.9562)^2 + (0.3)^2 - 2(0.9562\times 0.3\times\cos 60^o)} = 0.8470\ \text{m}$$

$$\text{and } \sin\theta = \left(\frac{0.3}{0.8470}\right)\times\sin 60^o = 0.3067 \Rightarrow \theta = \sin^{-1}(0.3067) = 17.86^o$$

Hence sum of vertical and horizontal forces can be given as:

$$T\cos 45^o - R\sin 17.86^o = 0$$

$$T\sin 45^o + 600 - R\cos 17.86^o = 0$$

Solving these equations simultaneously

$$R = \frac{600}{\cos 17.86^o - \sin 17.86^o} = 930.1\,N \quad and \quad T = \frac{930.1\times\sin 17.86^o}{\cos 45^o} = 403.4\,N$$

Equilibrium of a rigid body in three dimensions

Necessary and sufficient condition for equilibrium of a rigid body (in vector form) states:

$$\sum \mathrm{F} = 0 \qquad \text{and} \qquad \sum \mathrm{M} = 0 \qquad (3.10)$$

This **can be resolved into six scalar equations as follows:**

$$\sum F_x = 0 \qquad \sum F_y = 0 \qquad \sum F_z = 0$$
$$\sum M_x = 0 \qquad \sum M_y = 0 \qquad \sum M_z = 0 \qquad (3.11)$$

These 6 equations can be solved for no more than 6 unknowns which generally represent reactions at supports or connections

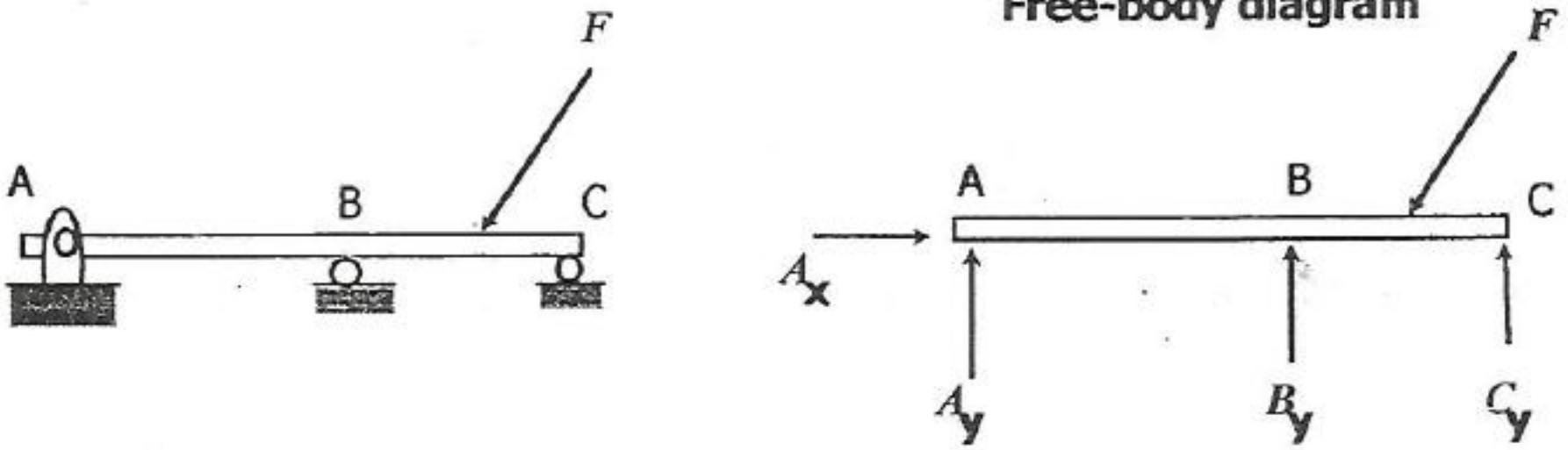


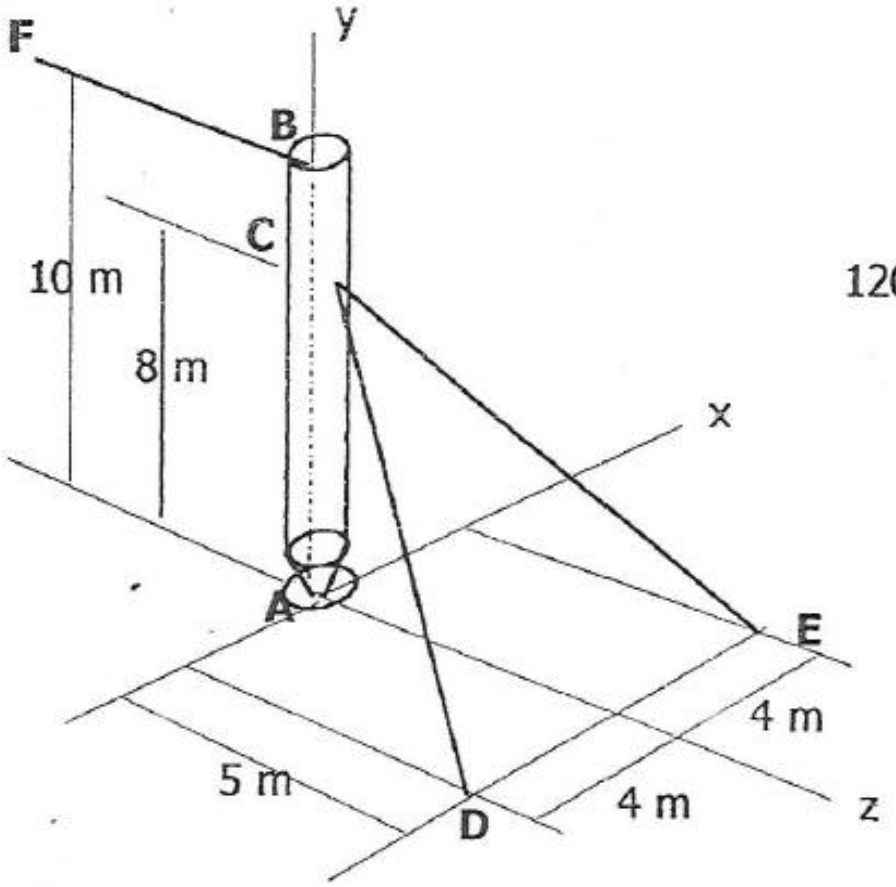
Fig. 3.8 Beam with 3 supports: a statically indeterminate problem

Equilibrium of a rigid body in three dimensions

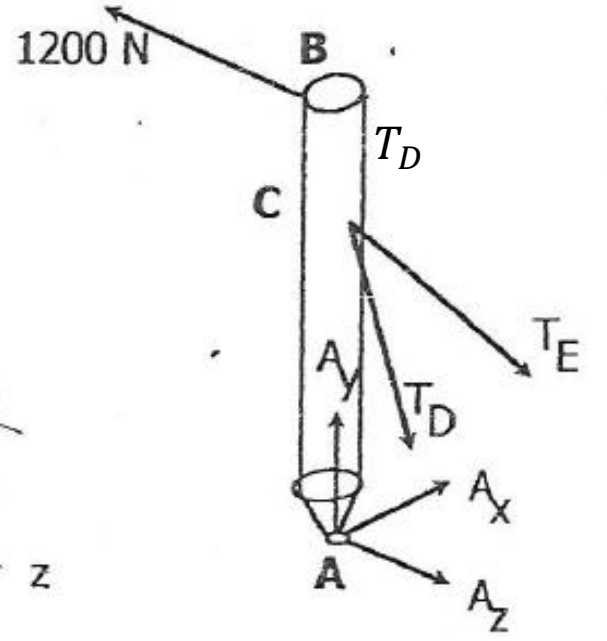
- **If all unknowns on the free-body diagram** can be determined from the equations of equilibrium, the **problem is said to be statically determinate**
- **Problems involving more unknowns than equations are referred to as statically indeterminate**
- Consider, for example, the 2-dimensional problem of a beam with 3 supports (See Fig. 3.8)
- Applications of the condition of equilibrium will yield only 3 equations whereas there are 4 unknowns Ax, Ay, By, and Cy.
- The statical indeterminacy is the result of redundancy of some of the constraints, i.e there are more constraints than is absolutely necessary to maintain equilibrium
- In order to obtain another equation, we would have to consider deformations of the beam at the supports
- Such considerations fall outside the scope of Rigid-body Mechanics and belong to the subject "Strength of Materials"

Example 3.4

The **10-metre mast shown in Fig. 3.9(a)** is **supported by a ball-and-socket joint at A and 2 cables CD and CE**. If the **mast supports a cable BF** which is horizontal and has tension of 1.2 kN, **determine the tension in cables CD and CE and the reactions at A.**



(a) mast



(b) Free-body diagram

Fig. 3.9 Body subjected to loads in 3-dimensions

SOLUTION

Each force is expressed in vector form as:

$$F = -1200\,\hat{k}$$

$$A = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$$T_D = \frac{\left(-4\hat{i} - 8\hat{j} + 5\hat{k}\right)}{\sqrt{4^2 + 8^2 + 5^2}} T_D = \left(-0.390\hat{i} - 0.781\hat{j} + 0.488\hat{k}\right) T_D$$

$$T_E = \frac{\left(4\hat{i} - 8\hat{j} + 5\hat{k}\right)}{\sqrt{4^2 + 8^2 + 5^2}} T_E = \left(0.390\hat{i} - 0.781\hat{j} + 0.488\hat{k}\right) T_E$$

Applying the equilibrium condition $\sum F = 0$:

$$F + A + T_D + T_E = 0 \quad \Rightarrow$$

$$-1200\hat{k} + A_x\hat{i} + A_y\hat{j} + A_z\hat{k} + \left(-0.390\hat{i} - 0.781\hat{j} + 0.488\hat{k}\right) T_D + \left(0.390\hat{i} - 0.781\hat{j} + 0.488\hat{k}\right) T_E$$

SOLUTION

Collecting the **i, j and k** terms together and equating to zero:

$$A_x - 0.390T_D + 0.390T_E = 0 \qquad (i)$$

$$A_y - 0.781T_D - 0.781T_E = 0 \qquad (ii)$$

$$-1200 + A_z + 0.488T_D + 0.488T_E = 0 \qquad (iii)$$

Applying the **condition ∑M = 0 and summing moments about point A**, we have

$$\begin{aligned} &10\,j \times (-1200k) \\ &+8\,j \times \left[(-0.390i - 0.781j + 0.488k)T_D\right] \\ &\quad 8\,j \times \left[(0.390i - 0.781j + 0.488k)T_E\right] = 0 \qquad \Rightarrow \\ &(-12000 + 3.904T_D + 3.904T_E)i + (3.120T_D - 3.120T_E)k = 0 \end{aligned}$$

Collecting terms in i, k and j, we have

$$-12000 + 3.904T_D + 3.904T_E = 0 \qquad (iv)$$

$$+3.120T_D - 3.120T_E = 0 \qquad (v)$$

SOLUTION

Solving equations (i) – (v) simultaneously:

$$T_D = T_E = \frac{1200}{2(3.904)} = 1536.9$$

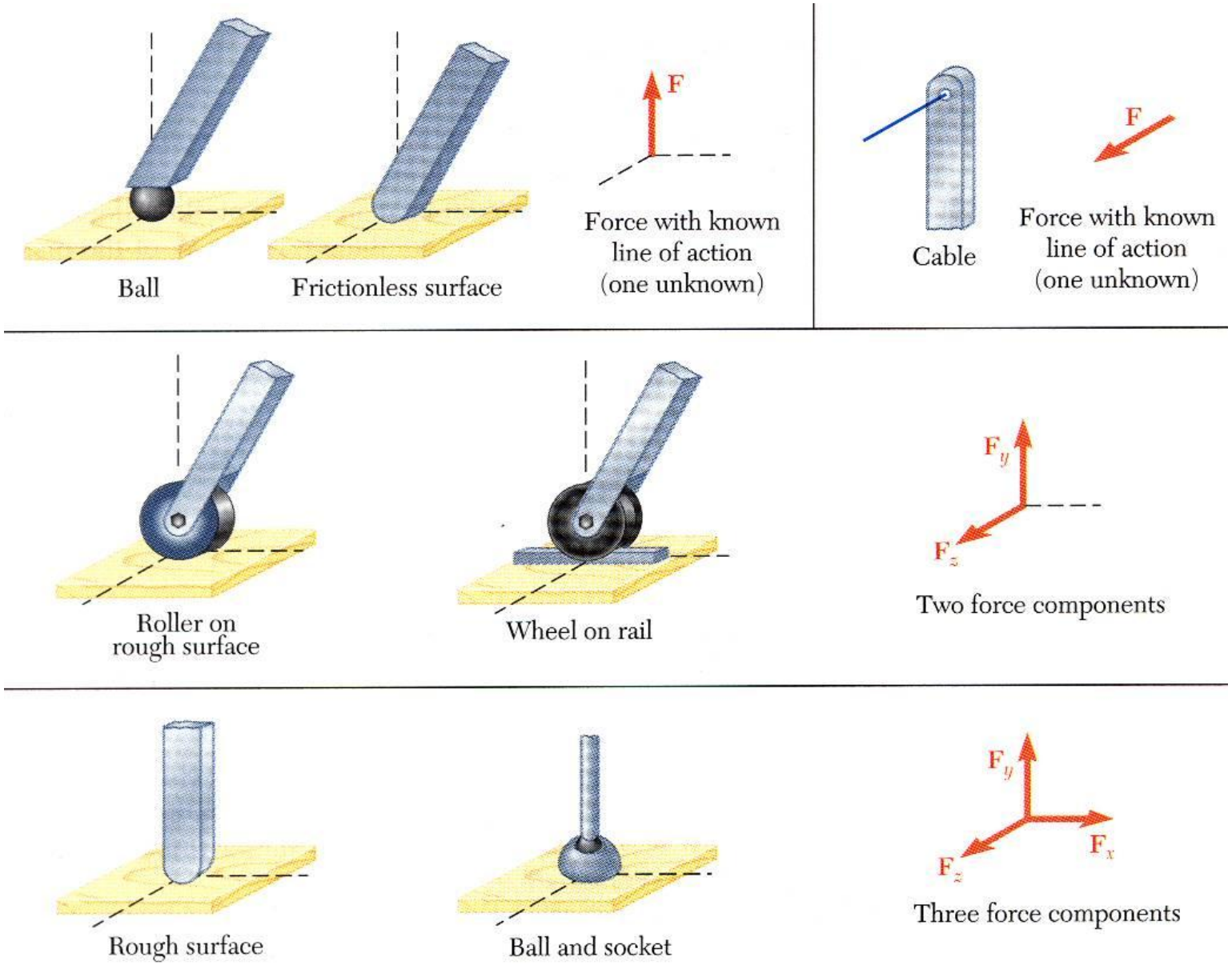
$$A_x = 0$$

$$A_y = 0.781 \times 1536.5 \times 2 = 2400.6\ \text{N}$$

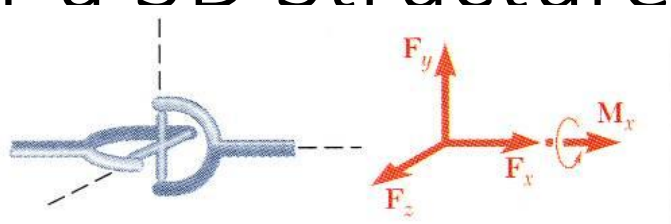
$$\text{and } A_z = 1200 - (0.488 \times 2 \times 1536.9) = -300\ \text{N}$$

- The **negative sign here shows** that the correct direction of **A_z is opposite to that shown on the free-body diagram**.
- Therefore **A_z = 300.0 N**

Reactions at Supports and Connections for a 3D Structure



Reactions at Supports and Connections for a 3D Structure

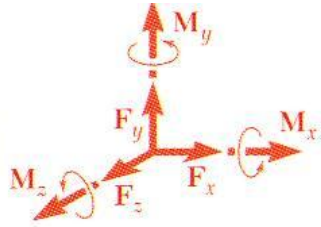


Universal joint

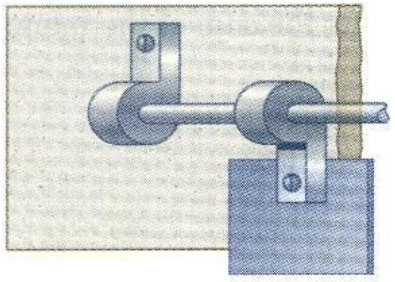
Three force components and one couple

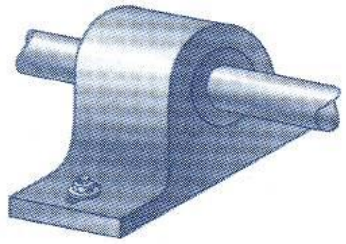


Fixed support

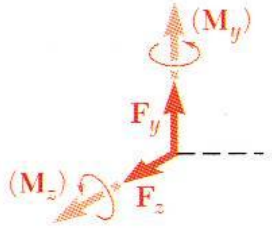


Three force components and three couples

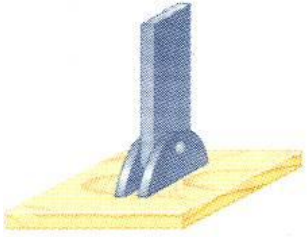




Hinge and bearing supporting radial load only

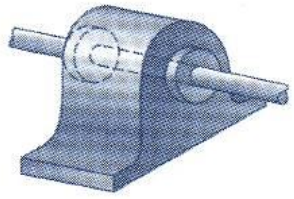


Two force components (and two couples)

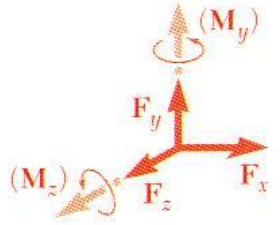


Pin and bracket





Hinge and bearing supporting axial thrust and radial load



Three force components (and two couples)

Thank You

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